Discussion Section 3
STAT 132 , Winter 2025

1. Suppose that $X_1, \ldots, X_n$ are a random sample from the Normal distribution with unknown mean μ and unknown variance σ^2. Find the maximum likelihood estimator for μ and σ^2?

2. A random variable X is modeled using the following PDF:

$$f(x|\theta) = \frac{\theta}{x^{\theta+1}} \quad \text{for } x > 1,$$

   where $\theta > 0$. Suppose you observe the random sample $X_1, \ldots, X_n$. What is the MLE of θ?

3. Suppose that $X_1, \ldots, X_n$ form a random sample from the uniform distribution on the interval $[\theta_1, \theta_2]$, where both θ_1 and θ_2 are unknown ($-\infty < \theta_1 < \theta_2 < \infty$). Find the MLE of θ_1 and θ_2.

4. Suppose that $X_1, \ldots, X_n$ form a random sample from an exponential distribution for which the value of the parameter λ is unknown. Determine the MLE of the median of the distribution.

5. Suppose that $X_1, \ldots, X_n$ are a random sample from the Normal distribution with unknown mean μ and unknown variance σ^2. Find the MLE of the 0.95 quantile of the distribution (i.e. the value of a such that $\mathbb{P}(X < a) = 0.95$).